

**Innovative Mathematics Olympiad
Model Paper 2025 – 2026**

Total Questions: 50

Time: 1.5 hr.

Total Marks: 100

Syllabus

Integers, Fractions and Decimals, Exponents and Powers, Rational Numbers, Data Handling, Simple Equations, Algebraic Expressions, Lines and Angles, Triangles and Its Properties, Congruence of Triangles, Comparing Quantities, Perimeter and Area, Visualising Solid Shapes.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

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| <ol style="list-style-type: none"> 1. A submarine is 120 m below sea level. It rises 45 m. Which integer represents its new position?
 (A) –165 (B) –75
 (C) 75 (D) 165 2. A bacteria population doubles every hour. Which exponential expression best represents the population after 6 hours, starting from one bacterium?
 (A) 2×6 (B) 6^2
 (C) 2^6 (D) 6^3 3. Statement I: Every proper fraction is less than 1.
 Statement II: In a proper fraction, the numerator is always smaller than the denominator.
 (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false | <ol style="list-style-type: none"> (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true 4. A student writes the number 5 as a fraction to show it is a rational number. Which of the following correctly represents this idea?
 (A) $\frac{5}{0}$ (B) $\frac{0}{5}$
 (C) $\frac{5}{1}$ (D) $\frac{1}{5}$ 5. A number is multiplied by itself 4 times. Which expression correctly represents this situation?
 (A) $4x$ (B) x^4
 (C) $x \times 4$ (D) x^3 6. If point A is 4 units left of zero and point B is 6 units right of zero, what is the integer value of $(A + B)$?
 (A) –10 (B) –2
 (C) 2 (D) 10 |
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7. Two straight rods cross each other in the shape of an "X". A mirror is placed exactly at the point of intersection. If one angle between the rods is 47° , the angle directly opposite to it is _____.
 (A) 47° (B) 90°
 (C) 133° (D) 180°
8. A point moves 12 units left, then 7 units right, then 15 units left from zero. Which integer represents its final position?
 (A) -20 (B) -10
 (C) -4 (D) 10
9. A triangular garden has angles in the ratio 3:4:5.
 If the shortest fence is placed opposite the smallest angle, which angle lies opposite the longest fence?
 (A) 3 parts angle (B) 4 parts angle
 (C) 5 parts angle (D) 6 parts angle
10. A lift goes down 12 floors from ground level, then goes up 7 floors, and then goes down 10 floors. Which integer represents its final position?
 (A) -15 (B) -5
 (C) 5 (D) 15
11. In an isosceles triangle, each equal angle is 40° . Find the angle between the equal sides.
 (A) 80° (B) 90°
 (C) 100° (D) 120°
12. A student spends $\frac{1}{3}$ of a day studying and $\frac{1}{6}$ of a day playing. What fraction of the day remains?
 (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
 (C) $\frac{1}{6}$ (D) $\frac{2}{3}$
13. A bicycle wheel rotates half a turn from point A to point B. The angle traced at the centre of the wheel is _____.
 (A) 90°
 (B) 180°
 (C) 270°
 (D) 360°
14. If one of the complementary angles is 35° , the other angle is _____.
 (A) 45° (B) 55°
 (C) 65° (D) 75°
15. If 15 workers can complete a job in 24 days, how many additional workers are required to complete the same job in 18 days?
 (A) 3 (B) 5
 (C) 7 (D) 10
16. Which of the following pairs must always be like fractions?
 (A) $\frac{3}{4}$ and $\frac{5}{6}$
 (B) $\frac{2}{9}$ and $\frac{7}{9}$
 (C) $\frac{4}{5}$ and $\frac{6}{7}$
 (D) $\frac{1}{3}$ and $\frac{3}{5}$
17. Without converting into decimals, which of the following rational numbers is the smallest?
 (A) $-\frac{7}{9}$ (B) $-\frac{5}{6}$
 (C) $-\frac{4}{9}$ (D) $-\frac{2}{3}$
18. A number is first raised to power 3 and the result is again raised to power 2. Which exponent represents the final result?
 (A) 5 (B) 6
 (C) 9 (D) 12

19. The information is given in which alternative is incorrectly matched?

Angle	Property
(A) Acute Angle	Measure is less than 90°
(B) Straight Angle	Measure is 180°
(C) Complementary Angles	Sum is 90°
(D) Vertically Opposite Angles	Sum is 180°

20. Which statement is always true?

- (A) A trinomial must contain three variables
- (B) A monomial can have more than one variable
- (C) Degree of a constant is 1
- (D) Like terms must have equal coefficients

21. A number written in standard form has a power of 10 that is positive. What does this indicate?

- (A) The number is a decimal less than 1
- (B) The number is negative
- (C) The number is very large
- (D) The number is zero

22. A fraction becomes $\frac{3}{4}$ when reduced to its simplest form. Which of the following could be the original fraction?

- (A) $\frac{6}{10}$
- (B) $\frac{9}{12}$
- (C) $\frac{12}{20}$
- (D) $\frac{15}{25}$

23. In a triangle, the longest side is **10 cm**. Which angle must be opposite this side?

- (A) Smallest angle

(B) Right angle only

(C) Largest angle

(D) Any angle

24. The price of an article is first increased by 10% and then again increased by 20%. The total increase in price is

- (A) 30%
- (B) 32%
- (C) 33%
- (D) 35%

25. Which option shows the correct descending order?

- (A) $\frac{1}{3}, \frac{1}{2}, \frac{2}{3}$
- (B) $\frac{2}{3}, \frac{1}{2}, \frac{1}{3}$
- (C) $\frac{1}{2}, \frac{2}{3}, \frac{1}{3}$
- (D) $\frac{1}{3}, \frac{2}{3}, \frac{1}{2}$

26. A square garden has the same perimeter as a rectangle of length 20 m and breadth 14m. What is the side length of the square?

- (A) 16 m
- (B) 17 m
- (C) 18 m
- (D) 19 m

27. An algebraic expression that contains three unlike terms after simplification is called a

- (A) Monomial
- (B) Binomial
- (C) Trinomial
- (D) Multinomial

28. The ratio of the lengths of two rods is 4:5. If the shorter rod is increased by 25% and the longer rod is decreased by 20%, what is the new ratio of their lengths?

- (A) 1:1
- (B) 5:4
- (C) 4:5
- (D) 3:4

29. A bar graph uses the scale 1 cm = 20 students. If a bar measures 4.5 cm, what number of students does it represent?

- (A) 70
- (B) 80
- (C) 90
- (D) 100

30. A fraction becomes equivalent to $\frac{5}{6}$ after multiplying both numerator and denominator by the same number. If the denominator of the new fraction is 48, what is the numerator?

- (A) 35 (B) 40
(C) 45 (D) 30

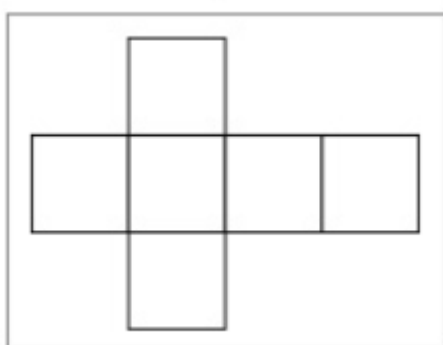
31. A shopkeeper mixes rice varieties A and B in the ratio 5:7. If the total mixture is 48 kg, how much rice of variety A is used?

- (A) 18 kg (B) 20 kg
(C) 24 kg (D) 28 kg

32. A solid has no edges and no vertices. Which statement about this solid is true?

- (A) It has flat faces
(B) It can be unfolded into a net
(C) It rolls smoothly in all directions
(D) It has exactly one face

33. When this net is folded, which solid will be formed?



- (A) Cuboid
(B) Cube
(C) Square pyramid
(D) Triangular prism

34. The table shows the shoe sizes of students in a class.

Shoe size	Number of students
5	7
6	10
7	6
8	4

Which shoe size would have the longest bar if this data is represented using a bar graph?

- (A) Size 5 (B) Size 6
(C) Size 7 (D) Size 8

35. A wire of length 88 m is used to make four identical square frames.

What is the area of each square frame?

- (A) 30.25 m^2 (B) 49 m^2
(C) 64.3 m^2 (D) 81 m^2

36. A cuboid has dimensions $4 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}$. If viewed from the side showing the maximum area, what is the area of that view?

- (A) 6 cm^2 (B) 8 cm^2
(C) 12 cm^2 (D) 24 cm^2

37. A bicycle covers 36 km in 3 hours. How far will it travel in 5 hours at the same speed?

- (A) 54 km (B) 60 km
(C) 72 km (D) 90 km

38. After simplification, how many terms are present in the expression

$$2x^2 + 3xy - xy - x^2 + 5?$$

- (A) 1 (B) 2
(C) 3 (D) 4

- 39.** The angles of a triangle are $(x+10)^\circ$, $(2x-10)^\circ$, and x° . Find the value of the largest angle.
 (A) 50° (B) 60°
 (C) 70° (D) 80°
- 40.** Two triangles are congruent. One triangle has angles 45° , 55° , and 80° . The corresponding largest angle of the other triangle is
 (A) 45° (B) 55°
 (C) 80° (D) 70°
- 41.** In $\triangle ABC$ and $\triangle PQR$, $AB = PQ$, $BC = QR$, and $\angle B = \angle Q$. Which congruence condition proves $\triangle ABC \cong \triangle PQR$?
 (A) SSS (B) SAS
 (C) ASA (D) RHS
- 42.** A bus travels x km every hour. Due to traffic, it travels 5 km less in the next hour. Which expression represents the total distance covered in two hours?
 (A) $2x - 5$ (B) $x - 5$
 (C) $x + 5$ (D) $2x + 5$
- 43.** The mean of 6 observations is 18. If one observation of value 30 is replaced by a new observation, the mean of the observations becomes 15. What is the value of the new observation?
 (A) 6 (B) 12
 (C) 18 (D) 24
- 44.** A number when multiplied by 4 and then reduced by 12 gives 28. Which equation represents the number?
 (A) $4x - 12 = 28$
 (B) $4(x - 12) = 28$
 (C) $x - 12 = 28$
 (D) $12x - 4 = 28$
- 45.** $\triangle ABC \cong \triangle PQR$. If $AB = 3x + 1$ and $PQ = 13$, find x .
 (A) 2 (B) 3
 (C) 4 (D) 5
- 46.** Two triangular road signs are made using two equal sides and the angle between them. Which congruence rule proves that the two signs are exactly the same in shape and size?
 (A) SSS (B) SAS
 (C) ASA (D) RHS
- 47.** If a number is increased by 9, the result becomes 23. Which equation correctly represents this situation?
 (A) $x + 23 = 9$ (B) $x - 9 = 23$
 (C) $x + 9 = 23$ (D) $23 + x = 9$
- 48.** Two triangles are congruent by RHS. Which of the following must be false?
 (A) Both triangles are right-angled
 (B) Hypotenuse is the longest side
 (C) All three sides are equal
 (D) One corresponding side is equal
- 49.** When a number is divided by 4, the result is 11. Which equation models this situation?
 (A) $4x = 11$ (B) $x \div 4 = 11$
 (C) $x - 4 = 11$ (D) $11 \div x = 4$
- 50.** The A rectangular garden has a length of 12 m and a breadth of 8 m. What is its perimeter?
 (A) 20 m (B) 40 m
 (C) 96 m (D) 48 m

KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) B | 2) C | 3) A | 4) C | 5) B | 6) C | 7) A |
| 8) A | 9) C | 10) A | 11) C | 12) A | 13) B | 14) B |
| 15) B | 16) B | 17) B | 18) B | 19) D | 20) B | 21) C |
| 22) B | 23) C | 24) B | 25) B | 26) B | 27) C | 28) B |
| 29) C | 30) B | 31) B | 32) C | 33) B | 34) B | 35) A |
| 36) C | 37) B | 38) C | 39) D | 40) C | 41) B | 42) A |
| 43) B | 44) A | 45) C | 46) B | 47) C | 48) C | 49) B |
| 50) B | | | | | | |